# Kernix 0.12.0 / Performance brief

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

**46** workloads

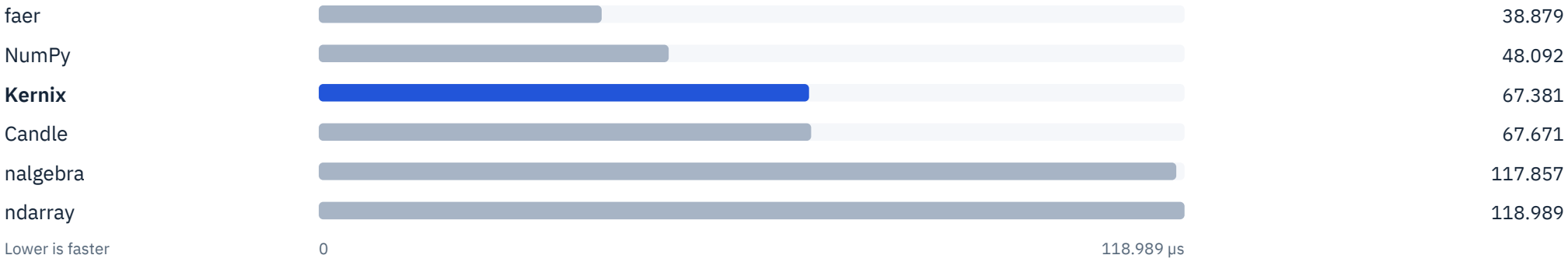
**152** observations

Source precision retained

## Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product




## How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

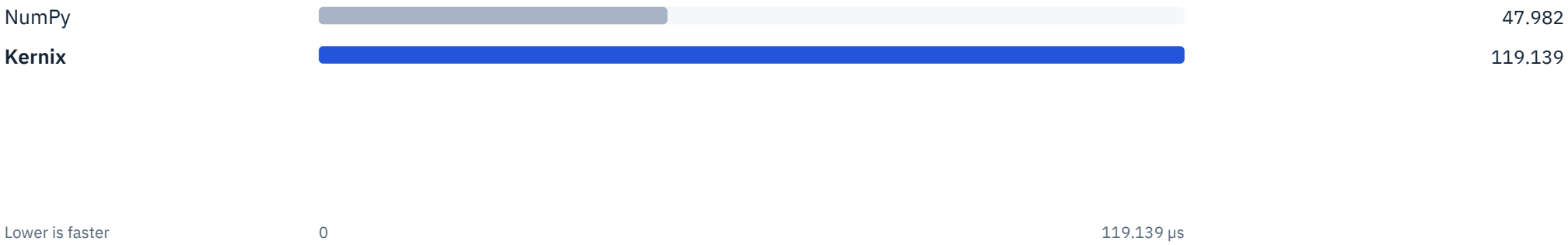
# Kernix 0.12.0 / Matrix multiplication

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Matrix multiplication · transposed view

Per-call median

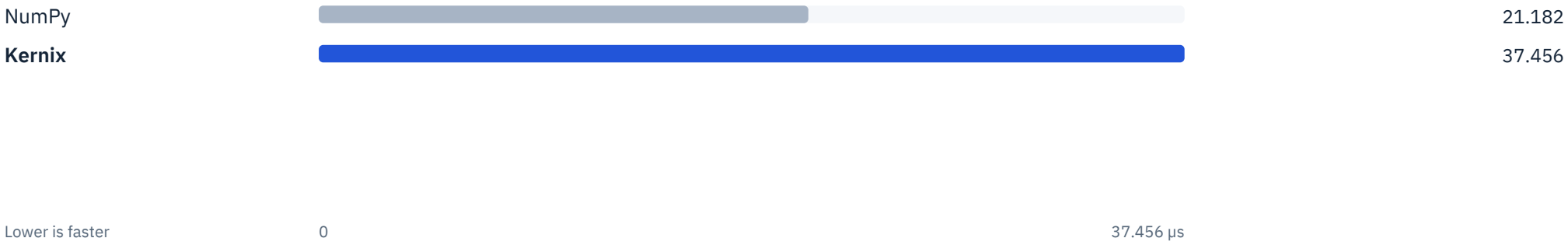
128 × 128 · 128 × 128 · f64 · 1 worker · borrowed transpose · allocated output · independent operands




## Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · borrowed transpose · allocated output · independent operands



# Kernix 0.12.0 / Matrix multiplication

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Independent matrix operands

Per-call median

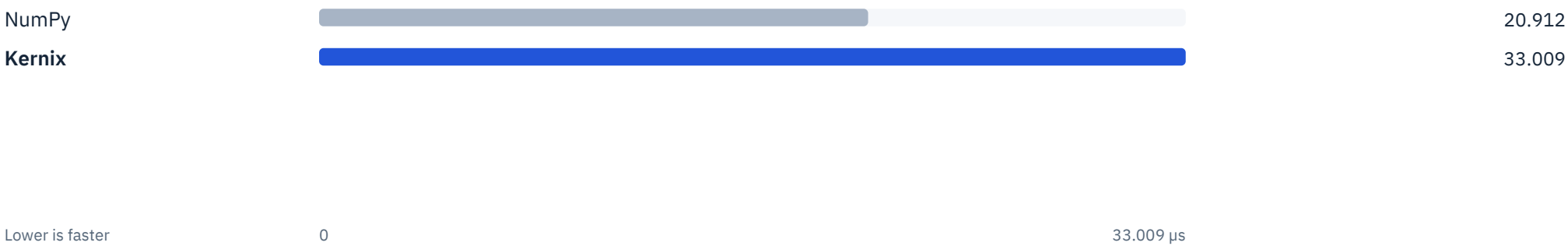
128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands




## Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



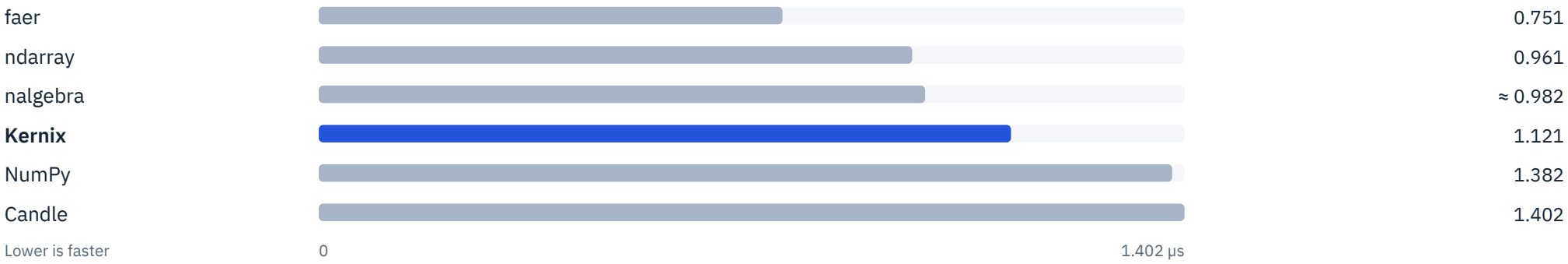
# Kernix 0.12.0 / Matrix multiplication

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Matrix multiplication

Per-call median

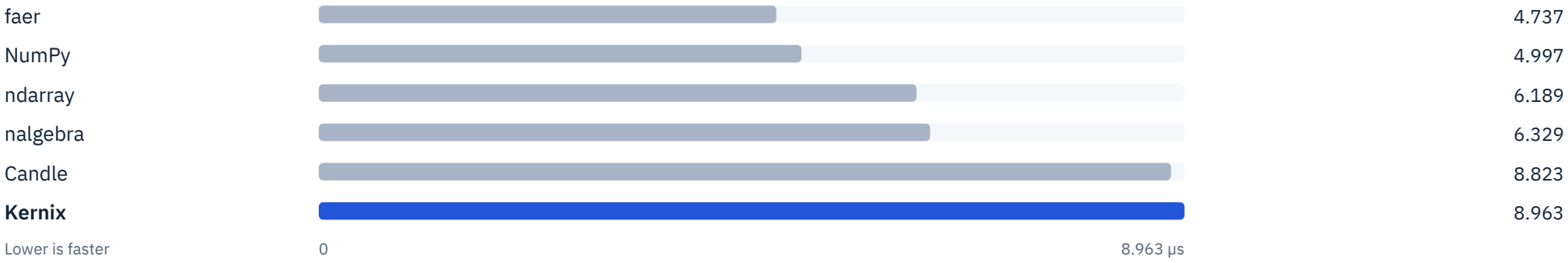
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product




## Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



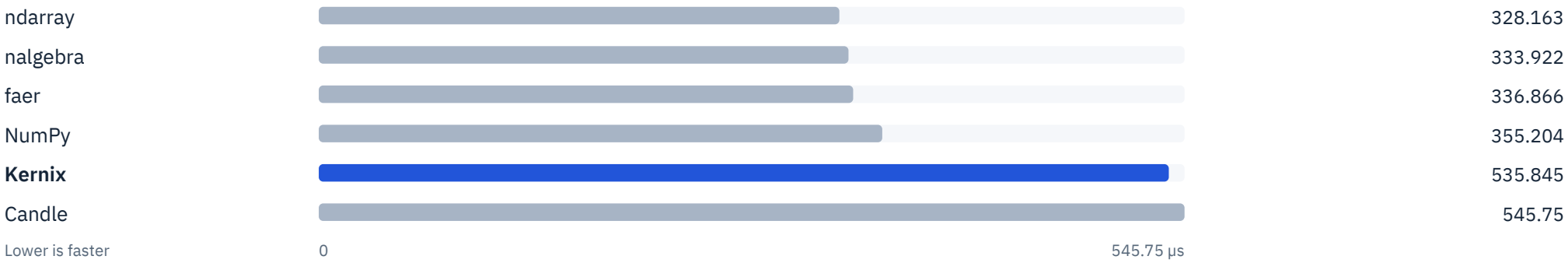
# Kernix 0.12.0 / Matrix multiplication

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product




## Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



# Kernix 0.12.0 / Matrix multiplication

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product




## Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



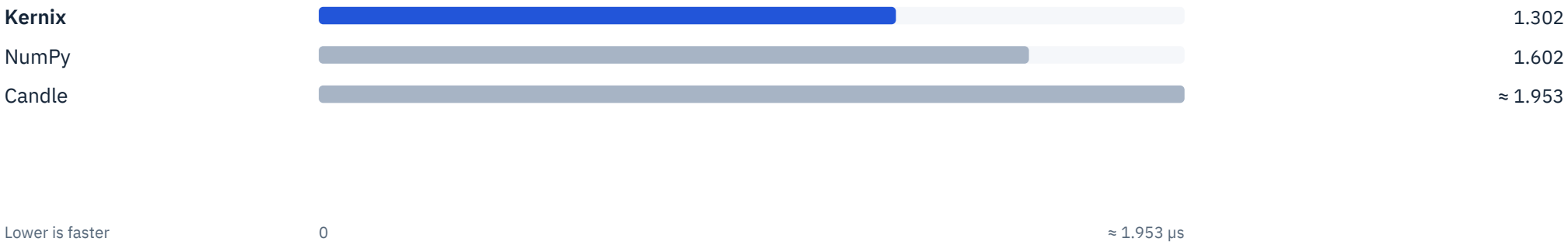
# Kernix 0.12.0 / Matrix multiplication

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Batched multiplication

Per-call median

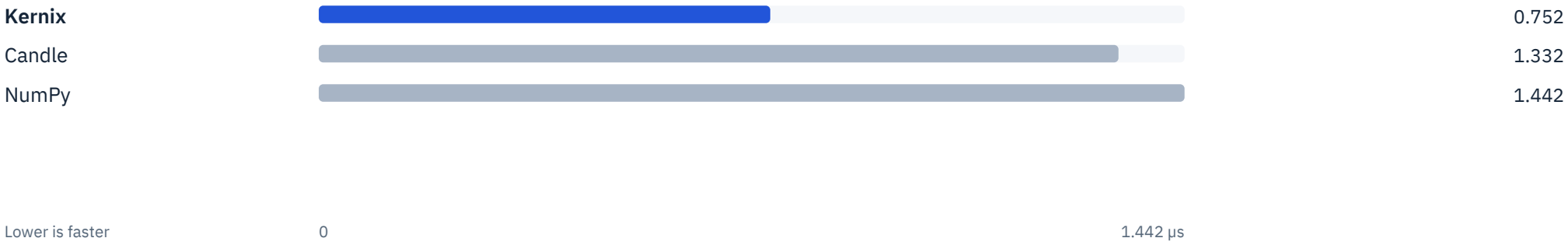
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches




## Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



# Kernix 0.12.0 / Explicit parallel execution

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

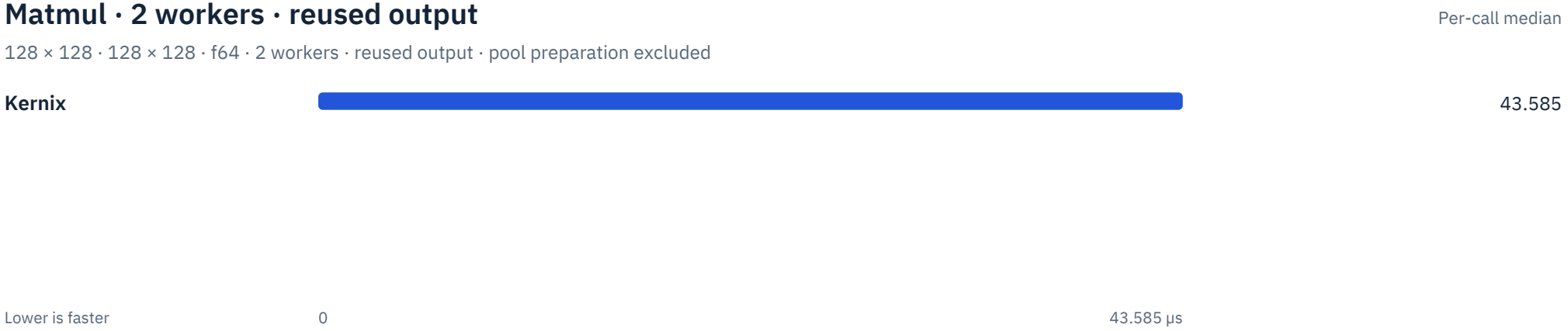
## Matmul · 2 workers · reused output

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded




## Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



# Kernix 0.12.0 / Explicit parallel execution

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded




## Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



# Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Cholesky solve · reused factor

Per-call median

128 × 128, 1 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded




## Cholesky solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



# Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Cholesky solve · reused factor

Per-call median

512 × 512, 128 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded




## LU solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



# Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

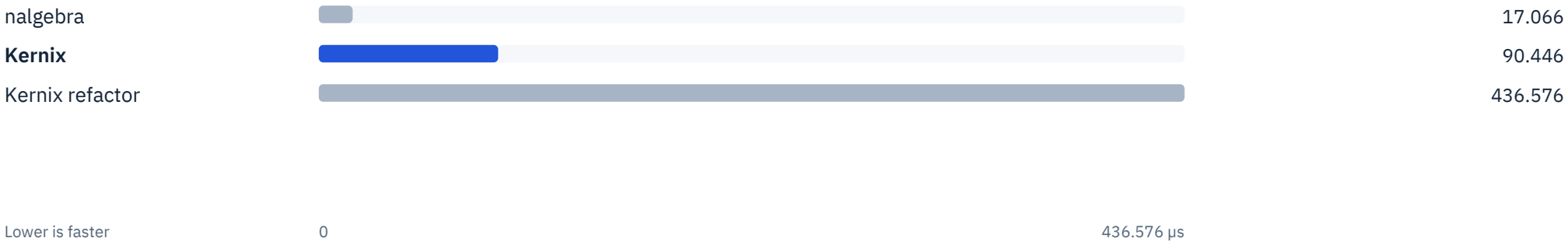
## Cholesky rank-one update

256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix




## Cholesky rank-one downdate

256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



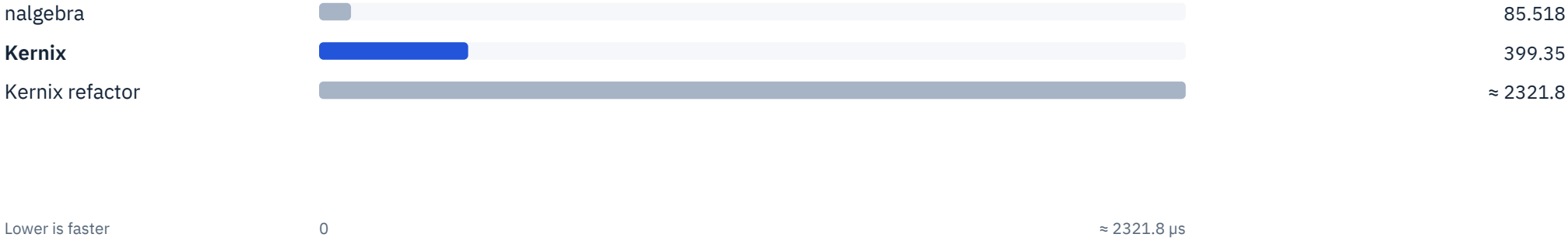
# Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Cholesky rank-one update

512 × 512 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix

Per-call median




## Thin QR

512 × 128 · f64 · 1 worker · allocated output

Per-call median



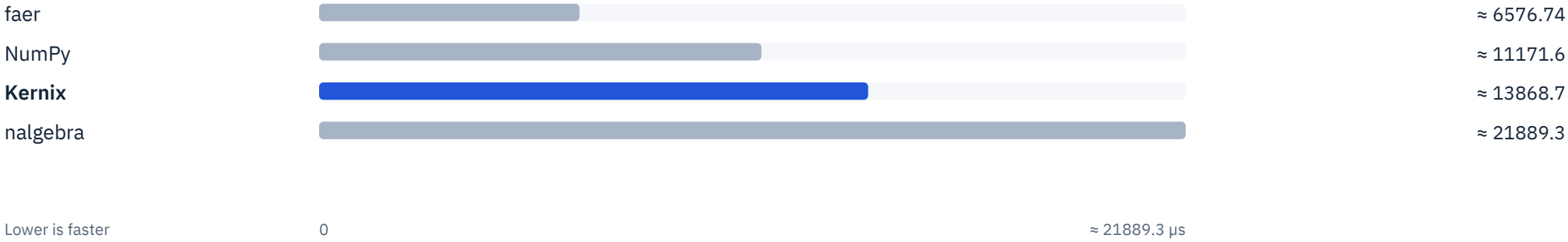
# Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Thin QR

1024 × 256 · f64 · 1 worker · allocated output

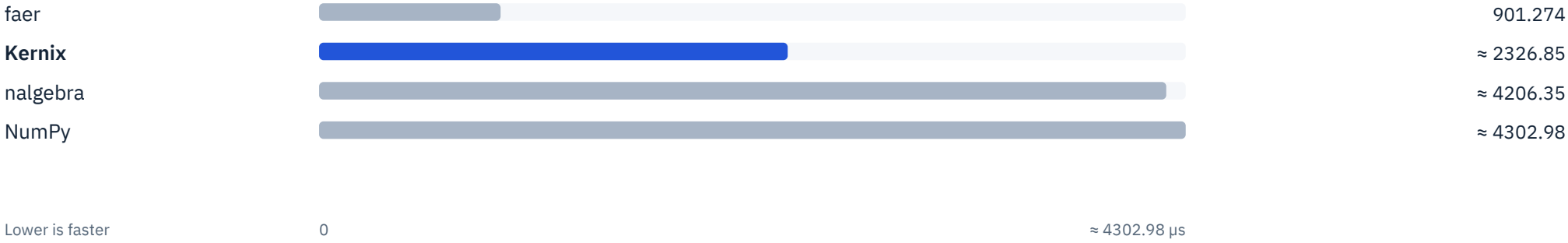
Per-call median




## Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output

Per-call median



# Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

Per-call median




## Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



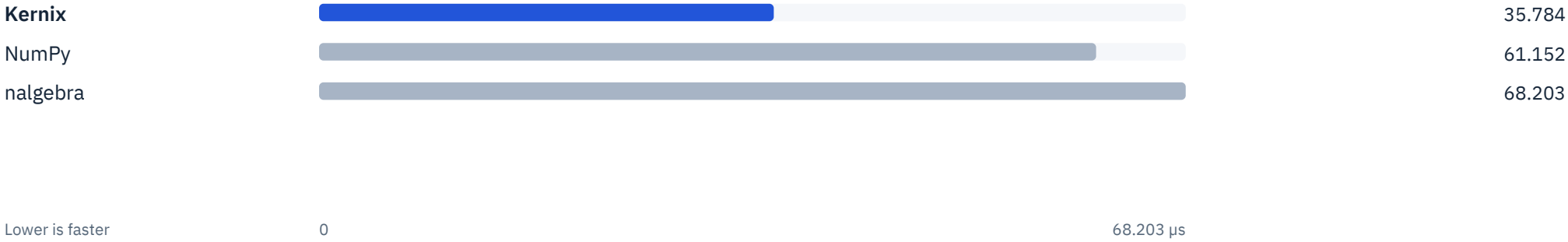
# Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Minimum-norm least squares

Per-call median

64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ




## Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



# Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Thin QR

96 × 32 · f64 · 1 worker · allocated output

Per-call median




## Thin QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



# Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Cholesky factorization

Per-call median

64 × 64 · f64 · 1 worker · allocated output




## Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



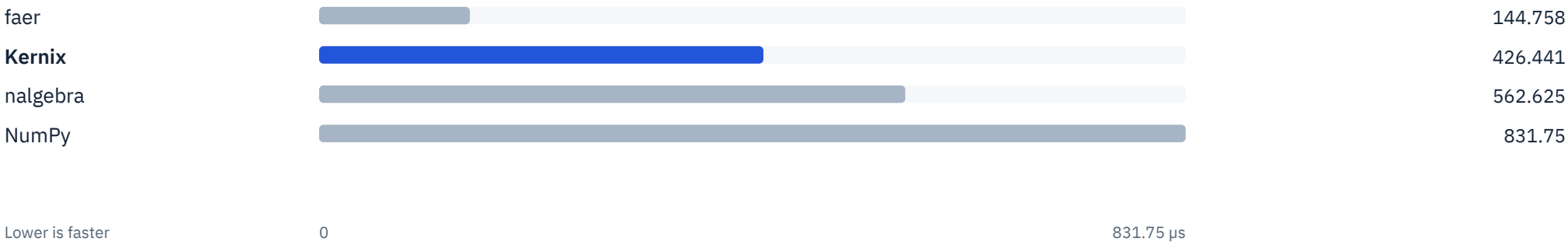
# Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output

Per-call median



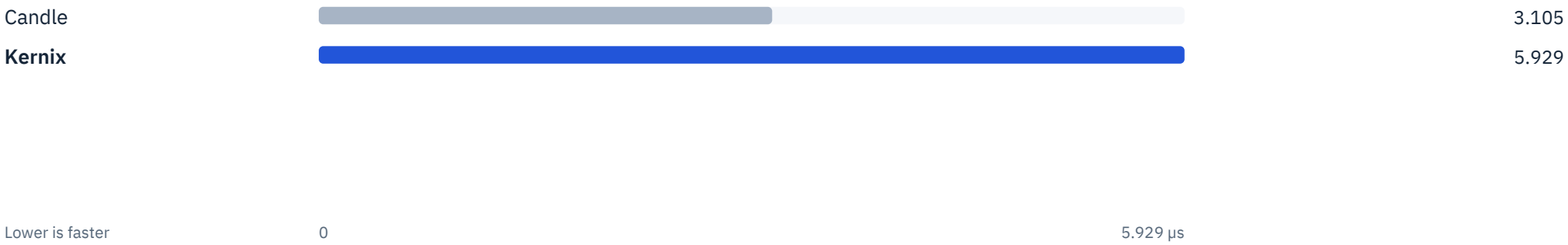
# Kernix 0.12.0 / Training & autodiff

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Shared-variable backward

Per-call median

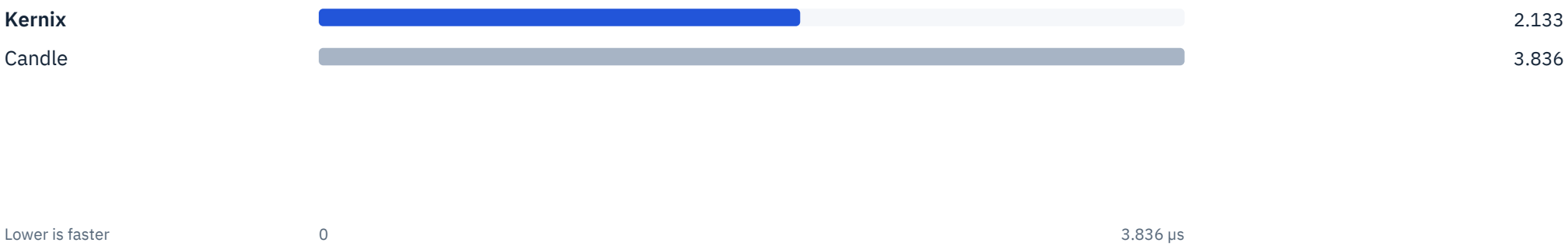
sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients




## Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



# Kernix 0.12.0 / Training & autodiff

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Model inference

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented




## Model recording + forward

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



# Kernix 0.12.0 / Training & autodiff

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients




## Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



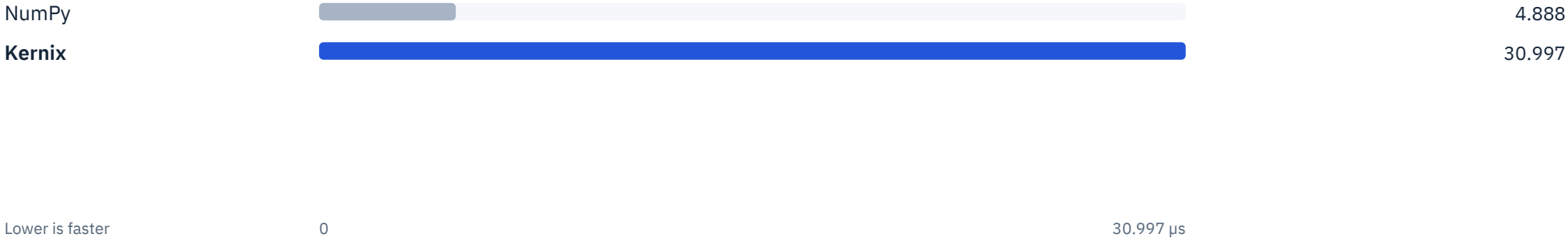
# Kernix 0.12.0 / Tensor contractions

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

Per-call median



# Kernix 0.12.0 / Statistics & activations

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Row softmax

256 × 64 · f64 · 1 worker · allocated output

Per-call median

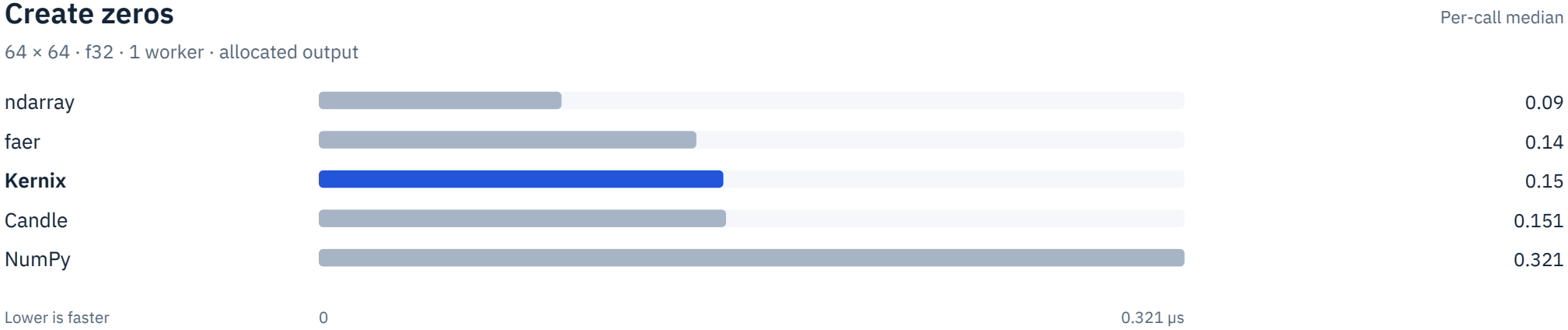


# Kernix 0.12.0 / Tensor construction

AMD EPYC 9V45 96-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

## Create zeros

64 × 64 · f32 · 1 worker · allocated output




## Create ones

64 × 64 · f32 · 1 worker · allocated output



# Version comparison / paired processes

AMD EPYC 9V45 96-Core Processor · 9 process pairs per workload · ratio below 1 means less time

## Cholesky solve · reused factor

128 × 128, 1 RHS · f64

**Timing direction unclear**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **1.0053625053625053** | 0.9724108658743633 – 1.0274227029389271 |

## Cholesky solve · reused factor

256 × 256, 64 RHS · f64

**Observed time reduction**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **0.22349955049075287** | 0.21900384782544965 – 0.22888203630554305 |

# Version comparison / paired processes

AMD EPYC 9V45 96-Core Processor · 9 process pairs per workload · ratio below 1 means less time

## LU solve · reused factor

256 × 256, 64 RHS · f64

**Observed time reduction**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **0.3114620050231865** | 0.30519189639903604 – 0.31553349932267266 |

## Thin QR

512 × 128 · f64

**Observed time reduction**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **0.6757057230163083** | 0.6720568966832867 – 0.6879233649014185 |

# Version comparison / paired processes

AMD EPYC 9V45 96-Core Processor · 9 process pairs per workload · ratio below 1 means less time

## Thin QR

1024 × 256 · f64

**Observed time reduction**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **0.5529138855671462** | 0.5445555682362697 – 0.556903078560186 |

## Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f64

**Timing direction unclear**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **0.9961673082199184** | 0.9935358919393151 – 1.0091649948805888 |

# Version comparison / paired processes

AMD EPYC 9V45 96-Core Processor · 9 process pairs per workload · ratio below 1 means less time

## Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f32

**Timing direction unclear**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **1.0005362505362505** | 0.9877411308939422 – 1.0212528876206006 |

## Thin QR

256 × 64 · f64

**Observed time reduction**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **0.9590911843475237** | 0.9536949327442831 – 0.9682737393033156 |

# Version comparison / paired processes

AMD EPYC 9V45 96-Core Processor · 9 process pairs per workload · ratio below 1 means less time

## Matrix multiplication

32 × 32 · 32 × 32 · f64

**Timing direction unclear**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **1.0008920606601248** | 0.9812834224598931 – 1.0268022181146026 |

## Matrix multiplication

128 × 128 · 128 × 128 · f64

**Timing direction unclear**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **0.9982135475107777** | 0.991413560752696 – 1.0239473217976731 |

# Version comparison / paired processes

AMD EPYC 9V45 96-Core Processor · 9 process pairs per workload · ratio below 1 means less time

## Matrix multiplication

128 × 128 · 128 × 128 · f32

**Timing direction unclear**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **1.0035971223021583** | 0.9870526408502392 – 1.0125065441219243 |

## Matrix multiplication

256 × 256 · 256 × 256 · f64

**Timing direction unclear**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **1.0012324316956442** | 0.9986771773084744 – 1.0135525631939524 |

# Version comparison / paired processes

AMD EPYC 9V45 96-Core Processor · 9 process pairs per workload · ratio below 1 means less time

## Matrix multiplication

256 × 256 · 256 × 256 · f32

**Timing direction unclear**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **1.001817487237072** | 0.9901220464195274 – 1.0192704641580272 |

## Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

**Observed time increase**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **1.0530631934394596** | 1.0391581008321096 – 1.0794833581718828 |

# Version comparison / paired processes

AMD EPYC 9V45 96-Core Processor · 9 process pairs per workload · ratio below 1 means less time

## Independent matrix operands

128 × 128 · 128 × 128 · f64

**Timing direction unclear**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **1.0054114825363514** | 0.9929526907242172 – 1.008696674109766 |

## Independent matrix operands

128 × 128 · 128 × 128 · f32

**Timing direction unclear**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **1.002366793881838** | 0.9940516705631677 – 1.0146177023358491 |

# Version comparison / paired processes

AMD EPYC 9V45 96-Core Processor · 9 process pairs per workload · ratio below 1 means less time

## Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

**Timing direction unclear**




| CURRENT / BASELINE | 1 | 95% INTERVAL | 0.9773071104387292 – 1.0152439024390243 |
|---|---|---|---|

## Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

**Timing direction unclear**




| CURRENT / BASELINE | 1 | 95% INTERVAL | 0.9986522911051213 – 1.015047879616963 |
|---|---|---|---|

# Version comparison / paired processes

AMD EPYC 9V45 96-Core Processor · 9 process pairs per workload · ratio below 1 means less time

## Cholesky factorization

128 × 128 · f64

**Timing direction unclear**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **0.9936608393422213** | 0.9854601575826497 – 1.009468378111521 |

## Cholesky factorization

256 × 256 · f64

**Timing direction unclear**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **1.0032612891221793** | 0.9983964429344863 – 1.0177890307996873 |

# Version comparison / paired processes

AMD EPYC 9V45 96-Core Processor · 9 process pairs per workload · ratio below 1 means less time

## Cholesky factorization

512 × 512 · f64

**Timing direction unclear**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **0.9988555053724515** | 0.9935237240137919 – 1.0168522189687361 |

## Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

**Timing direction unclear**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **1.0118995686103436** | 0.9831246342890579 – 1.029130116094862 |

# Version comparison / paired processes

AMD EPYC 9V45 96-Core Processor · 9 process pairs per workload · ratio below 1 means less time

## Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

**Timing direction unclear**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **1.0039262481234843** | 0.987681647914046 – 1.0154321417140948 |

## Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

**Timing direction unclear**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **1.0041587746026517** | 0.9931137845829789 – 1.0251479289940828 |

# Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication · transposed view / Kernix**<br>128 × 128 · 128 × 128 · f64 | 119139 | 119.139 |
| **Matrix multiplication · transposed view / NumPy**<br>128 × 128 · 128 × 128 · f64 | 47982 | 47.982 |
| **Matrix multiplication · transposed view / Kernix**<br>128 × 128 · 128 × 128 · f32 | 37456 | 37.456 |
| **Matrix multiplication · transposed view / NumPy**<br>128 × 128 · 128 × 128 · f32 | 21182 | 21.182 |
| **Independent matrix operands / Kernix**<br>128 × 128 · 128 × 128 · f64 | 67341 | 67.341 |
| **Independent matrix operands / NumPy**<br>128 × 128 · 128 × 128 · f64 | 47972 | 47.972 |
| **Independent matrix operands / Kernix**<br>128 × 128 · 128 × 128 · f32 | 33009 | 33.009 |
| **Independent matrix operands / NumPy**<br>128 × 128 · 128 × 128 · f32 | 20912 | 20.912 |
| **Matrix multiplication / Candle**<br>32 × 32 · 32 × 32 · f64 | 1402 | 1.402 |
| **Matrix multiplication / NumPy**<br>32 × 32 · 32 × 32 · f64 | 1382 | 1.382 |

# Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication / ndarray**<br>32 × 32 · 32 × 32 · f64 | 961 | 0.961 |
| **Matrix multiplication / faer**<br>32 × 32 · 32 × 32 · f64 | 751 | 0.751 |
| **Matrix multiplication / nalgebra**<br>32 × 32 · 32 × 32 · f64 | 982.0000000000001 | 0.9820000000000001 |
| **Matrix multiplication / Kernix**<br>32 × 32 · 32 × 32 · f64 | 1121 | 1.121 |
| **Matrix multiplication / faer**<br>64 × 64 · 64 × 64 · f64 | 4737 | 4.737 |
| **Matrix multiplication / nalgebra**<br>64 × 64 · 64 × 64 · f64 | 6329 | 6.329 |
| **Matrix multiplication / Candle**<br>64 × 64 · 64 × 64 · f64 | 8823 | 8.823 |
| **Matrix multiplication / ndarray**<br>64 × 64 · 64 × 64 · f64 | 6189 | 6.189 |
| **Matrix multiplication / NumPy**<br>64 × 64 · 64 × 64 · f64 | 4997 | 4.997 |
| **Matrix multiplication / Kernix**<br>64 × 64 · 64 × 64 · f64 | 8963 | 8.963 |

# Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication / Kernix**<br>128 × 128 · 128 × 128 · f64 | 67381 | 67.381 |
| **Matrix multiplication / nalgebra**<br>128 × 128 · 128 × 128 · f64 | 117857 | 117.857 |
| **Matrix multiplication / faer**<br>128 × 128 · 128 × 128 · f64 | 38879 | 38.879 |
| **Matrix multiplication / Candle**<br>128 × 128 · 128 × 128 · f64 | 67671 | 67.671 |
| **Matrix multiplication / ndarray**<br>128 × 128 · 128 × 128 · f64 | 118989 | 118.989 |
| **Matrix multiplication / NumPy**<br>128 × 128 · 128 × 128 · f64 | 48092 | 48.092 |
| **Matrix multiplication / faer**<br>256 × 256 · 256 × 256 · f64 | 336866 | 336.866 |
| **Matrix multiplication / nalgebra**<br>256 × 256 · 256 × 256 · f64 | 333922 | 333.922 |
| **Matrix multiplication / NumPy**<br>256 × 256 · 256 × 256 · f64 | 355204 | 355.204 |
| **Matrix multiplication / ndarray**<br>256 × 256 · 256 × 256 · f64 | 328163 | 328.163 |

# Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication / Candle**<br>256 × 256 · 256 × 256 · f64 | 545750 | 545.75 |
| **Matrix multiplication / Kernix**<br>256 × 256 · 256 × 256 · f64 | 535845 | 535.845 |
| **Matrix multiplication / Kernix**<br>64 × 64 · 64 × 64 · f32 | 4306 | 4.306 |
| **Matrix multiplication / Candle**<br>64 × 64 · 64 × 64 · f32 | 4387 | 4.387 |
| **Matrix multiplication / NumPy**<br>64 × 64 · 64 × 64 · f32 | 2894 | 2.894 |
| **Matrix multiplication / faer**<br>64 × 64 · 64 × 64 · f32 | 2333 | 2.333 |
| **Matrix multiplication / nalgebra**<br>64 × 64 · 64 × 64 · f32 | 3495 | 3.495 |
| **Matrix multiplication / ndarray**<br>64 × 64 · 64 × 64 · f32 | 3425 | 3.425 |
| **Matrix multiplication / Kernix**<br>128 × 128 · 128 × 128 · f32 | 34182 | 34.182 |
| **Matrix multiplication / ndarray**<br>128 × 128 · 128 × 128 · f32 | 40241 | 40.241 |

# Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication / faer**<br>128 × 128 · 128 × 128 · f32 | 19660 | 19.66 |
| **Matrix multiplication / nalgebra**<br>128 × 128 · 128 × 128 · f32 | 40181 | 40.181 |
| **Matrix multiplication / NumPy**<br>128 × 128 · 128 × 128 · f32 | 20420 | 20.42 |
| **Matrix multiplication / Candle**<br>128 × 128 · 128 × 128 · f32 | 33841 | 33.841 |
| **Matrix multiplication / nalgebra**<br>256 × 256 · 256 × 256 · f32 | 325859 | 325.859 |
| **Matrix multiplication / Kernix**<br>256 × 256 · 256 × 256 · f32 | 262855 | 262.855 |
| **Matrix multiplication / faer**<br>256 × 256 · 256 × 256 · f32 | 146280 | 146.28 |
| **Matrix multiplication / NumPy**<br>256 × 256 · 256 × 256 · f32 | 148143 | 148.143 |
| **Matrix multiplication / ndarray**<br>256 × 256 · 256 × 256 · f32 | 335454 | 335.454 |
| **Matrix multiplication / Candle**<br>256 × 256 · 256 × 256 · f32 | 262364 | 262.364 |

# Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Batched multiplication / NumPy**<br>4 × 16 × 32 · 1 × 32 × 16 · f64 | 1602 | 1.602 |
| **Batched multiplication / Kernix**<br>4 × 16 × 32 · 1 × 32 × 16 · f64 | 1302 | 1.302 |
| **Batched multiplication / Candle**<br>4 × 16 × 32 · 1 × 32 × 16 · f64 | 1953.0000000000002 | 1.9530000000000002 |
| **Batched multiplication / Kernix**<br>4 × 16 × 32 · 1 × 32 × 16 · f32 | 752 | 0.752 |
| **Batched multiplication / NumPy**<br>4 × 16 × 32 · 1 × 32 × 16 · f32 | 1442 | 1.442 |
| **Batched multiplication / Candle**<br>4 × 16 × 32 · 1 × 32 × 16 · f32 | 1332 | 1.332 |
| **Matmul · 2 workers · reused output / Kernix**<br>64 × 64 · 64 × 64 · f64 | 9344 | 9.344 |
| **Matmul · 2 workers · reused output / Kernix**<br>128 × 128 · 128 × 128 · f64 | 43585 | 43.585 |
| **Matmul · 2 workers · reused output / Kernix**<br>256 × 256 · 256 × 256 · f64 | 283426 | 283.426 |
| **Matmul · 2 workers · reused output / Kernix**<br>128 × 128 · 128 × 128 · f32 | 24807 | 24.807 |

# Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Cholesky solve · reused factor / faer**<br>128 × 128, 1 RHS · f64 | 3565 | 3.565 |
| **Cholesky solve · reused factor / nalgebra**<br>128 × 128, 1 RHS · f64 | 3645 | 3.645 |
| **Cholesky solve · reused factor / Kernix**<br>128 × 128, 1 RHS · f64 | 8923 | 8.923 |
| **Cholesky solve · reused factor / Kernix**<br>256 × 256, 64 RHS · f64 | 853993 | 853.993 |
| **Cholesky solve · reused factor / nalgebra**<br>256 × 256, 64 RHS · f64 | 725109 | 725.109 |
| **Cholesky solve · reused factor / faer**<br>256 × 256, 64 RHS · f64 | 122384.00000000001 | 122.38400000000001 |
| **Cholesky solve · reused factor / Kernix**<br>512 × 512, 128 RHS · f64 | 6608068 | 6608.068 |
| **Cholesky solve · reused factor / faer**<br>512 × 512, 128 RHS · f64 | 1113754 | 1113.754 |
| **Cholesky solve · reused factor / nalgebra**<br>512 × 512, 128 RHS · f64 | 5987768 | 5987.768 |
| **LU solve · reused factor / nalgebra**<br>256 × 256, 64 RHS · f64 | 675134 | 675.134 |

# Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **LU solve · reused factor / Kernix**<br>256 × 256, 64 RHS · f64 | 842977 | 842.977 |
| **LU solve · reused factor / faer**<br>256 × 256, 64 RHS · f64 | 187782 | 187.782 |
| **Cholesky rank-one update / nalgebra**<br>256 × 256 · f64 | 18007 | 18.007 |
| **Cholesky rank-one update / Kernix refactor**<br>256 × 256 · f64 | 435535 | 435.535 |
| **Cholesky rank-one update / Kernix**<br>256 × 256 · f64 | 92739 | 92.739 |
| **Cholesky rank-one downdate / nalgebra**<br>256 × 256 · f64 | 17066 | 17.066 |
| **Cholesky rank-one downdate / Kernix refactor**<br>256 × 256 · f64 | 436576 | 436.576 |
| **Cholesky rank-one downdate / Kernix**<br>256 × 256 · f64 | 90446 | 90.446 |
| **Cholesky rank-one update / Kernix refactor**<br>512 × 512 · f64 | 2321798 | 2321.798 |
| **Cholesky rank-one update / nalgebra**<br>512 × 512 · f64 | 85518 | 85.518 |

# Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Cholesky rank-one update / Kernix**<br>512 × 512 · f64 | 399350 | 399.35 |
| **Thin QR / Kernix**<br>512 × 128 · f64 | 2381458 | 2381.458 |
| **Thin QR / nalgebra**<br>512 × 128 · f64 | 2880959 | 2880.959 |
| **Thin QR / NumPy**<br>512 × 128 · f64 | 2084652 | 2084.652 |
| **Thin QR / faer**<br>512 × 128 · f64 | 1210359 | 1210.359 |
| **Thin QR / faer**<br>1024 × 256 · f64 | 6576743 | 6576.743 |
| **Thin QR / NumPy**<br>1024 × 256 · f64 | 11171564 | 11171.564 |
| **Thin QR / nalgebra**<br>1024 × 256 · f64 | 21889282 | 21889.282 |
| **Thin QR / Kernix**<br>1024 × 256 · f64 | 13868677 | 13868.677 |
| **Cholesky factorization / NumPy**<br>512 × 512 · f64 | 4302975 | 4302.975 |

# Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Cholesky factorization / nalgebra**<br>512 × 512 · f64 | 4206351 | 4206.351 |
| **Cholesky factorization / faer**<br>512 × 512 · f64 | 901274 | 901.274 |
| **Cholesky factorization / Kernix**<br>512 × 512 · f64 | 2326847 | 2326.847 |
| **Column-pivoted QR / faer**<br>96 × 32 · f64 | 23425 | 23.425 |
| **Column-pivoted QR / Kernix**<br>96 × 32 · f64 | 46029 | 46.029 |
| **Column-pivoted QR / nalgebra**<br>96 × 32 · f64 | 60451 | 60.451 |
| **Column-pivoted QR / nalgebra**<br>256 × 64 · f64 | 599630 | 599.63 |
| **Column-pivoted QR / Kernix**<br>256 × 64 · f64 | 524989 | 524.989 |
| **Column-pivoted QR / faer**<br>256 × 64 · f64 | 217167 | 217.167 |
| **Minimum-norm least squares / Kernix**<br>64 × 32, rank 16, 4 RHS · f64 | 35784 | 35.784 |

# Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Minimum-norm least squares / nalgebra**<br>64 × 32, rank 16, 4 RHS · f64 | 68203 | 68.203 |
| **Minimum-norm least squares / NumPy**<br>64 × 32, rank 16, 4 RHS · f64 | 61152 | 61.152 |
| **Minimum-norm least squares / nalgebra**<br>32 × 64, rank 32, 4 RHS · f64 | 125458 | 125.458 |
| **Minimum-norm least squares / Kernix**<br>32 × 64, rank 32, 4 RHS · f64 | 47371 | 47.371 |
| **Minimum-norm least squares / NumPy**<br>32 × 64, rank 32, 4 RHS · f64 | 115984 | 115.984 |
| **Thin QR / nalgebra**<br>96 × 32 · f64 | 33710 | 33.71 |
| **Thin QR / NumPy**<br>96 × 32 · f64 | 34822 | 34.822 |
| **Thin QR / faer**<br>96 × 32 · f64 | 28523 | 28.523 |
| **Thin QR / Kernix**<br>96 × 32 · f64 | 40851 | 40.851 |
| **Thin QR / NumPy**<br>256 × 64 · f64 | 314573 | 314.573 |

# Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Thin QR / nalgebra**<br>256 × 64 · f64 | 350937 | 350.937 |
| **Thin QR / faer**<br>256 × 64 · f64 | 233611 | 233.611 |
| **Thin QR / Kernix**<br>256 × 64 · f64 | 488283.99999999994 | 488.28399999999994 |
| **Cholesky factorization / faer**<br>64 × 64 · f64 | 6920 | 6.92 |
| **Cholesky factorization / Kernix**<br>64 × 64 · f64 | 16113.999999999998 | 16.113999999999998 |
| **Cholesky factorization / nalgebra**<br>64 × 64 · f64 | 14312 | 14.312 |
| **Cholesky factorization / NumPy**<br>64 × 64 · f64 | 10436 | 10.436 |
| **Cholesky factorization / faer**<br>128 × 128 · f64 | 27211 | 27.211 |
| **Cholesky factorization / NumPy**<br>128 × 128 · f64 | 88853 | 88.853 |
| **Cholesky factorization / Kernix**<br>128 × 128 · f64 | 97536 | 97.536 |

# Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Cholesky factorization / nalgebra**<br>128 × 128 · f64 | 84276 | 84.276 |
| **Cholesky factorization / Kernix**<br>256 × 256 · f64 | 426441 | 426.441 |
| **Cholesky factorization / nalgebra**<br>256 × 256 · f64 | 562625 | 562.625 |
| **Cholesky factorization / NumPy**<br>256 × 256 · f64 | 831750 | 831.75 |
| **Cholesky factorization / faer**<br>256 × 256 · f64 | 144758 | 144.758 |
| **Shared-variable backward / Candle**<br>sum(X * X), 64 × 32; prepared graph · f64 | 3105 | 3.105 |
| **Shared-variable backward / Kernix**<br>sum(X * X), 64 × 32; prepared graph · f64 | 5929 | 5.929 |
| **Record + backward / Candle**<br>sum(A · W), 32 × 16 · 16 × 8 · f64 | 3836 | 3.836 |
| **Record + backward / Kernix**<br>sum(A · W), 32 × 16 · 16 × 8 · f64 | 2133 | 2.133 |
| **Model inference / Candle**<br>batched normalized/sliced MLP · f64 | 2924 | 2.924 |

# Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Model inference / Kernix**<br>batched normalized/sliced MLP · f64 | 1933 | 1.933 |
| **Model recording + forward / Kernix**<br>batched normalized/sliced MLP · f64 | 2443 | 2.443 |
| **Model recording + forward / Candle**<br>batched normalized/sliced MLP · f64 | 4277 | 4.277 |
| **Model backward / Kernix**<br>prepared graph → fresh gradients · f64 | 3095 | 3.095 |
| **Model backward / Candle**<br>prepared graph → fresh gradients · f64 | 12739 | 12.739 |
| **Parameter update / Kernix**<br>transactional SGD, fixed gradients · f64 | 541 | 0.541 |
| **Coalesced tensor contraction / Kernix**<br>16 × 32 × 8 · 32 × 8 × 16 · f64 | 30997 | 30.997 |
| **Coalesced tensor contraction / NumPy**<br>16 × 32 × 8 · 32 × 8 × 16 · f64 | 4888 | 4.888 |
| **Row softmax / Kernix**<br>256 × 64 · f64 | 68864 | 68.864 |
| **Row softmax / ndarray**<br>256 × 64 · f64 | 42865 | 42.865 |

# Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Row softmax / NumPy**<br>256 × 64 · f64 | 44747 | 44.747 |
| **Row softmax / Candle**<br>256 × 64 · f64 | 44346 | 44.346 |
| **Create zeros / NumPy**<br>64 × 64 · f32 | 321 | 0.321 |
| **Create zeros / ndarray**<br>64 × 64 · f32 | 90 | 0.09 |
| **Create zeros / Kernix**<br>64 × 64 · f32 | 150 | 0.15 |
| **Create zeros / faer**<br>64 × 64 · f32 | 140 | 0.14 |
| **Create zeros / Candle**<br>64 × 64 · f32 | 151 | 0.151 |
| **Create ones / faer**<br>64 × 64 · f32 | 180 | 0.18 |
| **Create ones / Candle**<br>64 × 64 · f32 | 241 | 0.241 |
| **Create ones / NumPy**<br>64 × 64 · f32 | 1252 | 1.252 |

# Recorded precision / exact latency table

AMD EPYC 9V45 96-Core Processor · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Create ones / ndarray**<br>64 × 64 · f32 | 150 | 0.15 |
| **Create ones / Kernix**<br>64 × 64 · f32 | 220 | 0.22 |